

MUSTANG BEACH, TX, USA

WMO: 722008

Lat: 27.812N

Lon: 97.089W

Elev: 5

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 12995

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	36.7	40.6	24.1	18.3	46.2	28.5	22.5	48.1	28.4	56.3	26.2	52.5	15.8	0	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	90.7	80.6	90.2	80.6	89.6	80.6	84.1	86.8	83.0	86.4	82.2	85.9	11.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
83.7	176.4	85.0	82.0	166.4	84.2	81.3	162.7	84.0	48.4	84.6	46.6	85.7	46.1	87.1	88.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.7	21.1	19.0	33.6	93.6	3.1	3.3	31.4	96.0	29.6	97.9	27.9	99.7	25.7	102.1		
			29.7	84.0	3.5	1.8	27.1	85.3	25.1	86.4	23.1	87.4	20.6	88.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.9	56.8	60.0	65.9	72.8	78.2	83.6	84.5	85.2	83.1	77.1	66.8	60.0
	DBStd	11.66	7.67	8.29	6.75	4.95	3.85	2.16	1.75	1.86	3.46	5.80	8.71	7.94
	HDD50	79	31	22	4	0	0	0	0	0	0	0	7	15
	HDD65	791	264	167	71	9	0	0	0	0	0	0	83	192
	CDD50	8438	243	302	497	684	875	1008	1070	1090	992	842	510	325
	CDD65	3675	11	27	98	243	411	558	605	625	542	381	137	37
	CDH74	37322	5	25	103	957	3438	6705	7582	8055	6425	3471	534	22
	CDH80	13499	0	3	7	101	615	2554	3181	3642	2572	791	33	0
(14) Wind	WSAvg	10.4	10.0	10.2	10.5	11.8	12.1	11.1	10.5	9.3	9.8	10.0	10.1	9.7
(15) Precipitation	PrecAvg	34.3	2.1	2.1	2.6	2.1	3.3	2.6	3.1	2.0	5.9	4.2	2.8	1.7
	PrecMax	50.0	6.3	5.3	5.4	8.6	12.4	8.9	13.8	7.2	16.6	14.1	10.0	6.0
	PrecMin	18.2	0.0	0.1	0.1	0.2	0.0	0.0	0.0	0.0	1.3	0.0	0.0	0.3
	PrecStd	9.5	1.8	1.7	1.7	2.0	3.3	2.2	3.9	1.8	3.7	3.9	2.7	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.4	77.9	80.7	88.0	87.8	90.4	90.8	91.4	93.2	88.7	82.5	76.7
		MCWB	64.3	64.9	70.5	64.5	77.9	79.2	81.5	81.5	78.3	78.5	75.6	68.3
	2%	DB	71.7	73.1	76.3	82.0	85.6	89.8	90.2	90.8	90.6	87.3	81.0	73.4
		MCWB	65.6	67.8	71.4	74.4	78.9	79.9	81.3	81.3	79.9	78.5	74.8	69.8
	5%	DB	69.6	71.6	74.7	80.9	84.2	88.3	89.6	90.2	89.8	85.5	79.1	72.4
		MCWB	64.2	67.1	70.8	75.1	78.3	79.9	81.1	81.2	80.0	77.3	73.6	69.0
	10%	DB	66.8	69.7	73.0	79.0	83.2	87.9	88.3	89.6	88.3	83.9	77.2	70.9
		MCWB	63.7	66.1	69.8	73.9	77.9	79.8	80.5	81.1	79.3	76.4	72.8	67.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.0	71.5	75.5	81.2	84.0	84.9	85.3	83.8	85.7	83.2	78.8	73.3
		MCDB	71.8	73.5	76.2	81.2	84.2	86.6	87.1	88.8	87.4	84.5	80.7	74.2
	2%	WB	68.1	70.0	73.5	79.0	81.5	82.4	84.0	83.1	83.4	81.3	77.0	72.0
		MCDB	69.3	71.8	74.7	79.9	82.2	85.4	86.5	88.1	86.1	83.1	79.4	73.0
	5%	WB	66.3	68.5	72.5	77.2	80.6	81.8	82.5	82.4	81.9	79.7	75.2	70.2
		MCDB	67.6	70.1	73.7	79.2	82.0	85.1	86.0	87.4	85.9	82.9	77.6	71.4
	10%	WB	64.5	66.9	71.4	75.4	79.3	81.2	81.8	81.9	80.9	78.3	73.6	68.8
		MCDB	65.7	68.3	72.7	77.4	81.6	84.8	85.8	86.9	85.6	82.7	76.5	70.2
(35) Mean Daily Temperature Range	5% DB	MDBR	11.8	11.5	9.7	9.2	8.1	7.6	7.7	8.2	9.0	10.0	11.1	12.0
		MCDBR	13.3	11.8	9.1	10.0	8.3	8.1	8.2	8.7	9.6	8.9	9.5	11.6
	5% WB	MCWBR	10.9	8.9	6.7	7.0	4.2	3.6	3.5	3.4	4.5	5.4	6.8	9.6
		MCWBR	11.6	10.5	8.0	8.1	7.4	7.4	8.0	8.1	9.0	8.9	10.2	11.1
(40) Clear-Sky Solar Irradiance	taub		0.363	0.370	0.388	0.433	0.474	0.450	0.468	0.452	0.440	0.394	0.379	0.353
		taud	2.482	2.465	2.412	2.290	2.208	2.341	2.307	2.350	2.343	2.471	2.463	2.528
	Ebn at Noon		279	286	287	275	263	267	262	266	267	275	271	277
		Edh at Noon	30	33	37	43	46	40	42	40	39	32	30	28
(44) All-Sky Solar Radiation	RadAvg		947	1125	1447	1726	1943	2133	2096	2005	1657	1445	1082	853
	RadStd		104	184	153	164	132	110	173	108	140	145	114	114

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+146

Nomenclature: See separate page